

BARSARANI NAYAK

Generative AI & Agentic AI Developer

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PROFESSIONAL SUMMARY

Generative AI Developer specializing in LLM-powered applications, Retrieval-Augmented Generation (RAG) pipelines, and multi-agent AI systems. Skilled in LangChain, LangGraph, OpenAI API, Ollama, Hugging Face, and vector databases (ChromaDB), with strong expertise in prompt engineering and agent orchestration. Proven ability to translate business requirements into scalable, reliable AI solutions that automate workflows, reduce hallucinations, and improve user experience, backed by four end-to-end projects with automated testing and human-in-the-loop controls.

TECHNICAL SKILLS

Generative AI & LLMs: LangChain, LangGraph, OpenAI API, Groq, Ollama, Hugging Face, Transformers, Prompt Engineering, RAG, LLM Application Development

Agentic AI: AI Agents, Multi-Agent Systems, Agent Orchestration, Model Context Protocol (MCP), MCP Servers, Human-in-the-Loop Workflows, AI Workflow Automation

AI / ML: Machine Learning, Deep Learning, Natural Language Processing (NLP), Embeddings, Semantic Search

Vector Search & Data: ChromaDB, Vector Databases, LlamaIndex, MySQL, SQLite (FTS5), NumPy, Pandas

Backend & APIs: Python, FastAPI, Flask, Django, REST APIs, RESTful Web Services, API Development, Pydantic

Tools & Practices: Git, GitHub, Jira, VS Code, Google Antigravity, pytest, SDLC

PROJECT EXPERIENCE

Multi-Agent AI Research System

May 2026 – Present

Tech Stack: Python, LangGraph, Groq, ChromaDB, Scrapy, Requests, BeautifulSoup, Pydantic, pytest | [GitHub Repository](#)

- Developed a LangGraph-based multi-agent research system that automates source discovery, summarization, citation generation, trend analysis, and source-quality evaluation.
- Built a configurable crawling pipeline (Scrapy, Requests, BeautifulSoup) with same-domain traversal, robots.txt compliance, adjustable crawl depth, and local document storage.
- Implemented RAG-based trend analysis with ChromaDB to ground new findings in previously collected sources.
- Designed reflection workflows covering accuracy assessment, bias analysis, confidence scoring, and configurable human-review thresholds.
- Added human-in-the-loop approve/reject checkpoints with resumable state, so interrupted runs continue without reprocessing completed stages.
- Improved fault tolerance with LLM/crawler fallbacks, malformed-response recovery, CLI controls, and automated pytest coverage.

Smart Campus Multi-Agent System

Apr 2026 – Present

Tech Stack: Python, FastAPI, LangGraph, React 18, SQLite, FTS5, Groq, HTTPX | [GitHub Repository](#)

- Built a LangGraph multi-agent assistant that classifies and routes natural-language requests for class schedules, library services, campus locations, and reminders.
- Developed FastAPI REST APIs and a responsive React 18 frontend supporting chat, library search, reminders, student preferences, staff approvals, and usage analytics.
- Implemented persistent storage in SQLite with FTS5 full-text search for fast library catalog retrieval.
- Designed staff workflows for book reservations and loan-extension approvals, plus automated class reminder management.
- Integrated Groq LLM inference via HTTPX for query parsing and response generation, with deterministic fallback logic for offline or API-unavailable scenarios.
- Wrote automated tests covering agent routing, retrieval, API workflows, and core application logic.

Autonomous AI Software Engineering Team

Feb 2026 – Aug 2026

Tech Stack: Python, LLMs, Multi-Agent Systems, Jira, GitHub, QA Automation, Workflow Orchestration | [GitHub Repository](#)

- Architected an autonomous multi-agent workflow that converts natural-language requirements into backlogs, technical designs, implementation tasks, code reviews, and QA validation.
- Designed role-based AI agents with defined responsibilities, orchestration logic, validation steps, and approval checkpoints across the SDLC.

- Integrated Jira and GitHub to automate issue creation, task tracking, repository changes, and code review with end-to-end traceability.
- Improved reliability with bounded retries, validation checks, exception handling, and human-in-the-loop approval gates.
- Built live and offline execution modes so workflows can be developed, tested, and debugged without external LLM APIs.

Help Desk RAG Chatbot

Oct 2025 – Dec 2025

Tech Stack: Python, LangChain, ChromaDB, Groq, Llama 3.3, Hugging Face, Sentence Transformers, Streamlit, pytest | [GitHub Repository](#)

- Engineered a RAG chatbot with LangChain, ChromaDB, Groq, and Llama 3.3 that delivers grounded customer-support answers from internal knowledge-base documentation.
- Built an end-to-end ingestion and semantic search pipeline: Markdown parsing, text chunking, Sentence Transformer embeddings, and persistent ChromaDB storage.
- Implemented multi-turn conversational memory and session management to preserve context across follow-up troubleshooting queries.
- Reduced unsupported or hallucinated answers through source attribution, retrieval validation, and confidence checks.
- Designed an automated escalation workflow that creates JSON support tickets when documentation is missing or model confidence is low.
- Delivered CLI and Streamlit interfaces with pytest suites covering chatbot, retrieval, conversation, and escalation flows.

EDUCATION

Master of Science (M.Sc.) in Mathematics

2022 – 2024

Manav Rachna University, Faridabad | CGPA: 8.60

Bachelor of Science (B.Sc.) in Mathematics

2019 – 2022

Lingayas University, Faridabad | CGPA: 9.55